

Student GPA Prediction based on Performance using Different Machine Learning Models.

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Abstract — This study looks at predicting student GPA using machine learning models based on academic and non-academic data. Random Forest, Gradient Boosting, AdaBoost, Linear Regression, Ridge, Lasso, SVR, K-Nearest Neighbors, and XGBoost are among the models evaluated. R2 and Mean Squared Error (MSE) are used to assess performance. The most accurate models were Ridge and Linear Regression, which suggested a significant linear connection between variables and GPA. Because of its simplicity and accuracy, the research identifies Ridge Regression as the best model for predicting GPA, providing insightful information that may help educational institutions support student interventions.

Keywords — GPA Prediction, Machine Learning, Ridge Regression, Linear Regression, Random Forest, Support Vector Regression, XGBoost, Educational Data, Student Performance.

I. INTRODUCTION

It is well known that students have some indicators of performance, for example the Grade Point Average (GPA) which is useful for assessing performance in recent educational systems. GPA is one of the key performance indicators of students that are very critical in making decisions on further education, seeking internships, employment, scholarships and other benefits [2]. However, even though there are various identified ways of enhancing academic performance; these include academic input methods such as past grades and non-academic factors such as socio-economic status, attendance, and studying methods, predicting GPA has been a big challenge [4].

In today's world, where student outcomes can be complex, machine learning has been adopted in the Education system especially on demographic forecasting [1]. ML models help to investigate and analyze large sets of data, identify complicated structures and relationships among data, as well as predict future events from what has happened before [5]. This ability extends beyond simple statistical calculations towards providing a more advanced understanding of academic performance related areas.

To analyze the students' GPA prediction problem, this study addresses the effectiveness of various machine learning models namely, random Forest (RF), Gradient Boosting (GB), AdaBoost, Linear Regression (LR), Ridge, Lasso, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), and XGBoost. Each of these models has its own merits for its capability of handling various data types and variable combinations. The objective of the study is to evaluate such models with a view of employment in determining the students' performance matrices while providing valuable information to the concerned institutions and their teachers.

This research has more significance that goes well beyond enhancing GPA prediction accuracy. It provides a model in which educational institutions can develop their capacity to provide students' support systems. It is possible to address the problems by finding the troublemakers early on and using strategies such as academic support, tutoring, and college scholarships which will ultimately raise their grades [3]. This research therefore contributes to the knowledge and practice of the ways of managing education systems and students' success strategies.

II. LITERATURE REVIEW

Machine learning techniques have gained significant attention in educational settings, particularly for predicting student performance. Several studies have explored and compared different algorithms to determine their effectiveness in forecasting student outcomes and supporting timely interventions.

H. Al-Shehri et al. [1] conducted a comparative analysis of Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) models, using a dataset from Portuguese secondary schools to predict student grades. Their findings revealed that SVM slightly outperformed KNN, with correlation coefficients of 0.96 and 0.95, respectively, demonstrating the high predictive power of both models in academic contexts.

Similarly, H.M.R. Hasan et al. [2] utilized machine learning models such as Decision Tree and KNN to predict student performance with an impressive accuracy of 94.44%. By analyzing academic data from 1,170 students, their study focused on identifying at-risk students for early intervention. They proposed that future research could incorporate psychological data to enhance prediction accuracy further.

Building on the theme of early prediction, N. Alangari and R. Alturki [3] explored 15 different machine learning algorithms to predict students' final GPA. Naive Bayes and Hoeffding Tree emerged as the best-performing models, achieving 91% accuracy. The study highlighted the critical role of early course performance in shaping final GPA, offering valuable insights into how early academic indicators can guide student interventions and support.

In contrast, A.S. Hashim et al. [4] found that Logistic Regression was the most accurate model for predicting student performance, with an accuracy of 68.7% for predicting final grades and 88.8% for pass/fail status. This research underscores the potential of machine learning to support educational institutions by flagging students who may require additional assistance to improve their academic outcomes.

Finally, B. Sekeroglu et al. [5] investigated the application of machine learning models such as Support Vector Regression (SVR), Long Short-Term Memory (LSTM), and Backpropagation (BP) to predict and classify student performance. SVR was the best predictor, while BP achieved the highest classification accuracy at 87.78%. This study highlights the growing importance of machine learning in analyzing educational data for more effective student performance predictions.

Collectively, these studies demonstrate that various machine learning models, including SVM, Naive Bayes, Logistic Regression, and SVR, are effective in predicting student performance, with the potential to improve academic outcomes through early identification of at-risk students.

III. METHODOLOGY

This analysis aims to build machine learning models to predict student GPA using various regression techniques, including ensemble methods and linear models. Below is the step-by-step methodology:

A. Data Collection and Preparation:

- The dataset is loaded into a pandas DataFrame.
- Irrelevant columns, such as "StudentID", are removed to clean the data.
- Categorical variables (e.g., "Gender", "Ethnicity") are encoded using Label Encoding, transforming them into numerical representations.

- Numerical variables (e.g., "Age", "StudyTimeWeekly") are standardized using StandardScaler to normalize their scales.

B. Feature Selection and Target Definition:

- The features (x) are defined by excluding the target variable ("GPA").
- The target variable (y) is the student's GPA.

C. Train-Test Split:

- The data is split into training (80%) and testing (20%) sets to evaluate model performance on unseen data.

D. Model Selection:

A variety of machine learning models are chosen, including:

- Ensemble Methods: Random Forest, Gradient Boosting, AdaBoost, XGBoost.
- Linear Models: Linear Regression, Ridge, Lasso.
- Other Regressors: Support Vector Regression (SVR), K-Nearest Neighbors (KNN).

Each model is hyperparameter-tuned using GridSearchCV, with 5-fold cross-validation to optimize for the best performance.

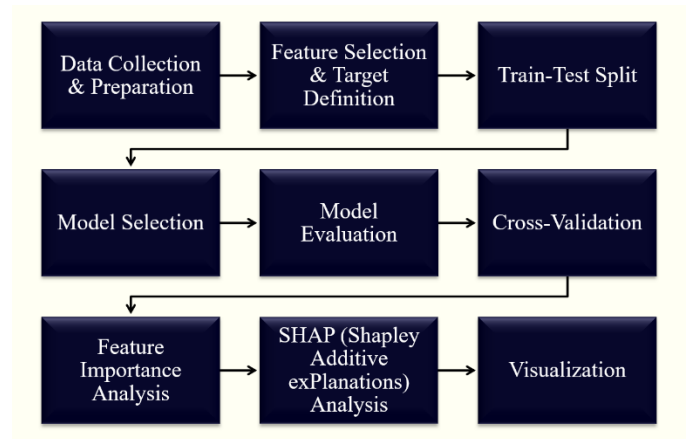


Figure 1: Process of Methodology

E. Model Evaluation:

Models are evaluated on the test set using metrics:

- R^2 Score: Measures the proportion of variance in GPA explained by the model.
- Mean Squared Error (MSE): Quantifies the prediction error.

Results are presented as a comparison of R^2 and MSE for each model.

F. Cross-Validation:

- The best-performing model is further validated using 5-fold cross-validation to assess its consistency across different data splits.

G. Feature Importance Analysis:

- For models like Random Forest, XGBoost, and Ridge, feature importance and coefficients are analyzed to understand which features contribute most to GPA predictions.

H. SHAP (SHapley Additive exPlanations) Analysis:

- SHAP values are computed for XGBoost to interpret the model's predictions, providing insight into how each feature impacts individual predictions.

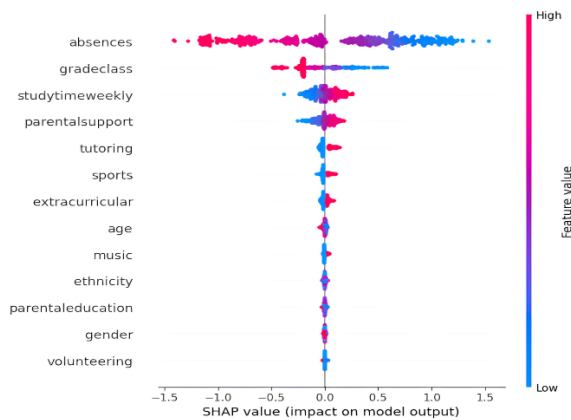


Figure 2: SHAP values Analysis

The SHAP summary plot visualizes the impact of individual features on predictions made by a machine learning model, likely a Random Forest, based on the code provided. The x-axis represents SHAP values, which indicate the contribution of each feature to the prediction. Positive SHAP values signify that a feature increases the predicted output, while negative values indicate a decrease. The y-axis lists the features in descending order of importance, with the most influential feature placed at the top.

In this plot, Absences emerges as the most significant feature, exhibiting a wide range of SHAP values. High absences (in red) generally lead to an increase in the predicted outcome, while low absences (in blue) tend to reduce it. Other notable features such as GradeClass and StudyTimeWeekly also show substantial influence on the model's predictions. Specifically, StudyTimeWeekly demonstrates a mostly positive effect as its value increases. The color gradient, which ranges from red (high feature value) to blue (low feature value), aids in understanding the relationship between feature values and their respective influence on predictions.

Features like Volunteering, Gender, and Ethnicity exhibit SHAP values close to zero, suggesting minimal contribution to the model's predictions. This plot provides a clear

representation of how different features, both numerical and categorical, affect the model's performance, with certain features showing significantly stronger influence than others.

I. Visualization:

- Results are visualized using bar plots for R^2 and MSE, scatter plots for actual vs. predicted GPA, residual plots, feature importance plots, and SHAP summary plots.

Here, we can see the actual GPA vs predicted GPA in scatter plot for every predicted model which we used.

The scatter plot shows the predicted GPA versus the actual GPA from a Lasso regression model. The red line represents the ideal line where predicted GPA perfectly matches the actual GPA, indicating good predictive accuracy.

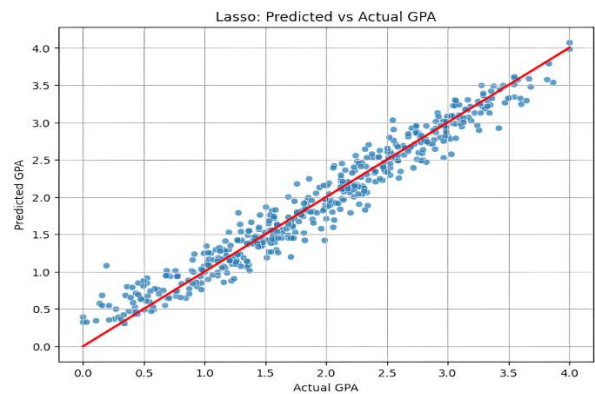


Figure 3: Scatter plot for LASSO

The scatter plot shows the predicted GPA versus the actual GPA from a Support Vector Regression (SVR) model. The red line represents the ideal scenario where predicted GPA matches the actual GPA, demonstrating strong predictive performance.

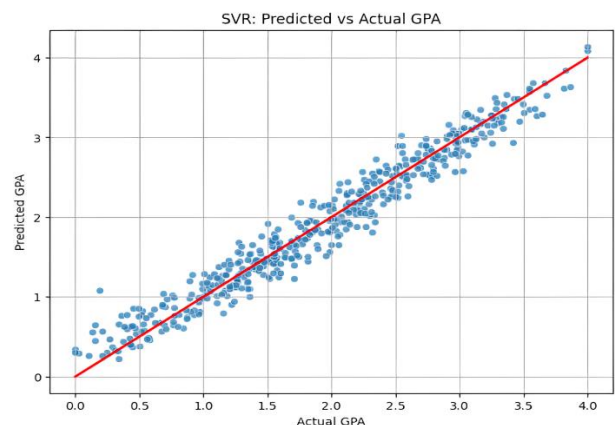


Figure 4: Scatter plot for SVR

The scatter plot illustrates the predicted GPA versus the actual GPA using a K-Nearest Neighbors (KNN) model. The red line signifies the perfect prediction line, showing the alignment between the predicted and actual GPA values.

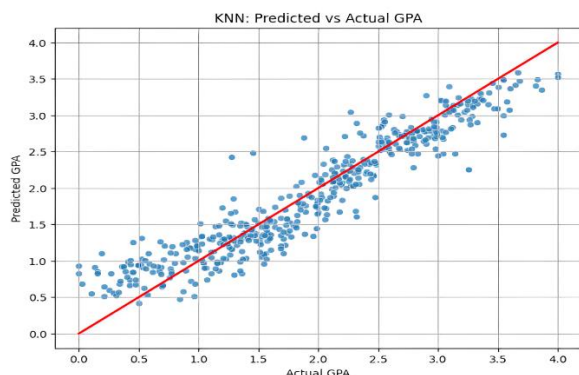


Figure 5: Scatter plot for KNN

The scatter plot visualizes the correlation between predicted and actual GPA values, suggesting a strong positive linear relationship. The red line, likely representing a perfect prediction, highlights the accuracy of the XGBoost model in predicting GPA.

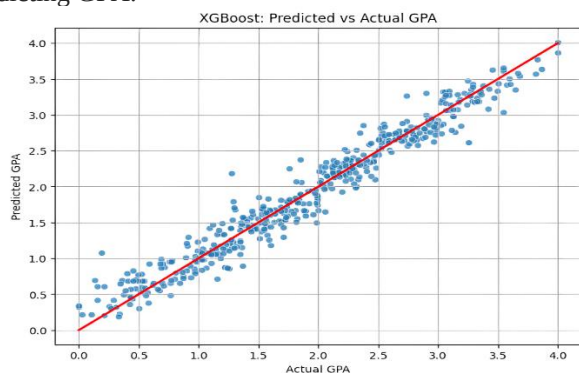


Figure 6: Scatter plot for XGBoost

The scatter plot shows the predicted versus actual GPA using a Random Forest model. The red line represents the perfect prediction line, indicating where predicted GPA would exactly match the actual GPA.

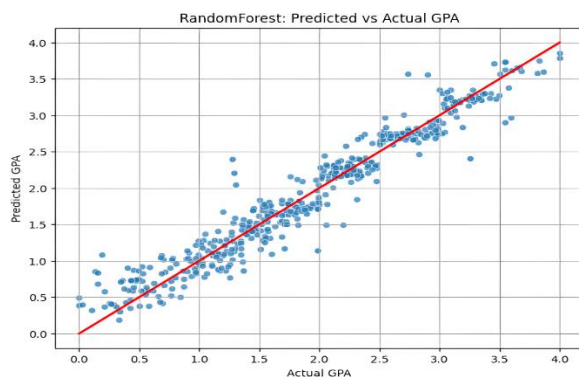


Figure 7: Scatter plot for RANDOMFOREST

The scatter plot compares the predicted GPA values from a Gradient Boosting model to the actual GPA values. The red line represents the ideal case where predicted values perfectly match the actual values.

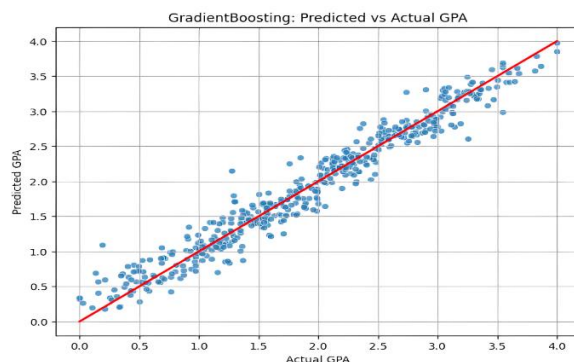


Figure 8: Scatter plot for GRADIENTBOOSTING

The scatter plot illustrates the relationship between predicted and actual GPA values using an AdaBoost model. The red line represents the ideal scenario where the predicted values perfectly align with the actual values.

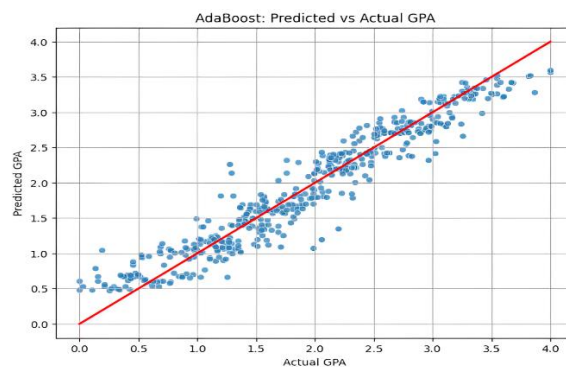


Figure 9: Scatter plot for AdaBoost

This scatter plot shows the relationship between predicted and actual GPA values using a Linear Regression model. The red line represents the perfect alignment between predicted and actual values, indicating the model's accuracy.

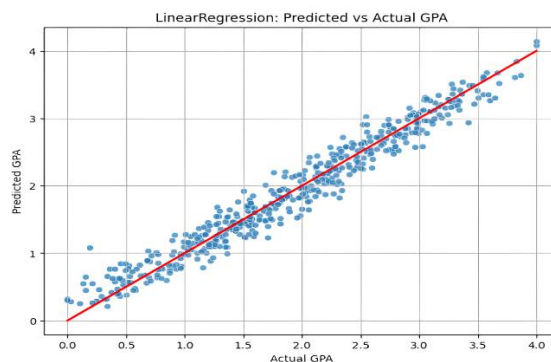


Figure 10: Scatter plot for LinearRegression

The scatter plot displays the predicted versus actual GPA using a Ridge regression model. The red line represents perfect predictions, where the predicted GPA matches the actual GPA exactly.

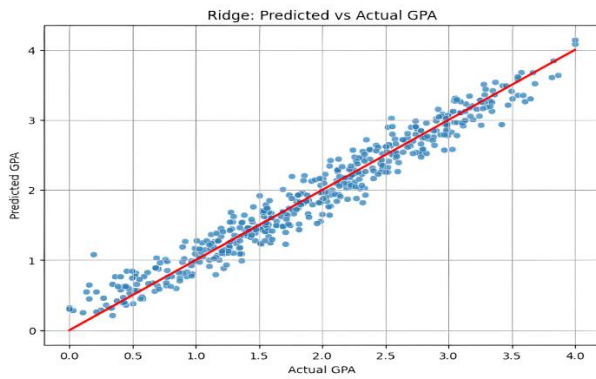


Figure 11: Scatter plot for RIDGE

The graphs display the predicted vs. actual GPA for various models, and all of them show predictions closely aligning with the actual values. This suggests that the models—whether linear or non-linear—are capturing the same underlying patterns in the data. The closeness between the results across different algorithms indicates that the relationship between features and GPA is highly linear, leading to similar performance across these models.

IV. RESULT

The table and graphs in the results section showcase a comparative analysis of various machine learning models based on their R^2 values, Mean Squared Error (MSE), and Mean Absolute Error (MAE), highlighting Ridge and Linear Regression as top performers in terms of accuracy, both achieving the lowest MSE and MAE. In contrast, K-Nearest Neighbors exhibited the lowest performance, with the highest MSE and MAE values.

Model Comparison:

Model	R^2 Values	MSE	MAE
Ridge	95.683363%	0.035696	0.150637
Linear Regression	95.683290%	0.035696	0.150639
SVR	95.659228%	0.035895	0.151026
XGBoost	95.576631%	0.036578	0.145224
Gradient Boosting	95.493605%	0.037265	0.146663
Lasso	95.467509%	0.037481	0.151654
Random Forest	93.785145%	0.051393	0.167903
AdaBoost	92.637405%	0.060884	0.192798
K-Nearest Neighbors	89.015029%	0.090838	0.233688

We can see the result in the graph also,

The bar chart compares the performance of 9 different Machine Learning models using the R^2 Score metric. All the models have very similar performance with R^2 Scores between 89% and 96%.

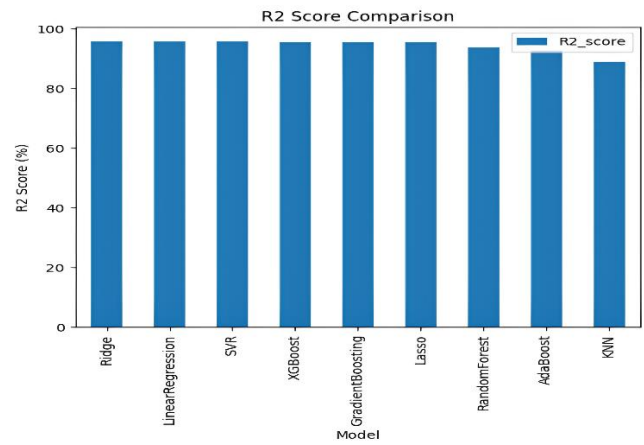


Figure 12: R2 Score Comparison

This bar chart displays the mean squared error (MSE) for different regression models. The KNN model has the highest MSE, indicating it performed the worst, while the Ridge model had the lowest MSE, performing the best.

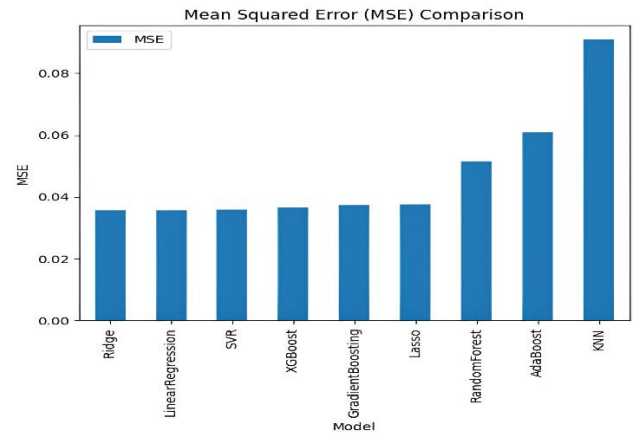


Figure 13: MSE Comparison

This bar chart displays the mean absolute error (MAE) for different regression models. The KNN model has the highest MAE, indicating it performed the worst, while the Ridge model had the lowest MAE, performing the best.

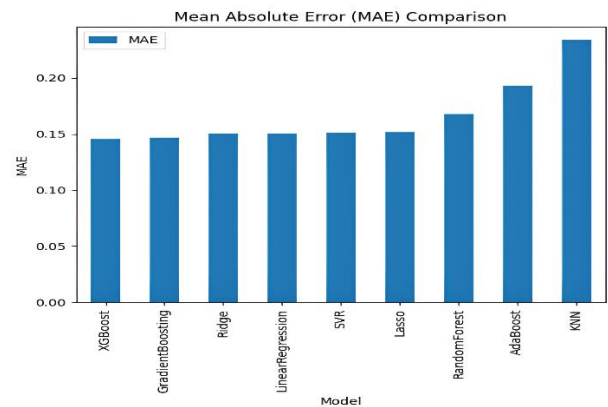


Figure 14: MAE Comparison

V. DISCUSSION

The study evaluated the performance of multiple regression models in predicting student GPA using R^2 scores, Mean Squared Error (MSE), and Mean Absolute Error (MAE). R^2 measures the model's ability to explain data variance, while MSE and MAE quantify prediction errors, with MAE reflecting the average absolute difference between predicted and actual values.

Ridge Regression and Linear Regression were the top performers, both achieving R^2 scores of 95.68%. This indicates their strong ability to explain data variance. The similarity in performance suggests that regularization in Ridge Regression doesn't significantly affect its accuracy. Both models also had the lowest MSE of 0.035696 and almost identical MAE values of 0.150637 and 0.150639, respectively, showcasing their high prediction accuracy and minimal deviation between predicted and actual GPA values.

Support Vector Regression (SVR) and XGBoost followed closely, with R^2 scores of 95.66% and 95.58%, respectively. These models handle nonlinear relationships effectively, explaining their strong results. Their MAE values were slightly higher, with SVR at 0.151026 and XGBoost at 0.145224, indicating a slightly higher error in individual predictions but still within an acceptable range.

Gradient Boosting, Lasso, and Random Forest also performed well, with R^2 scores ranging between 95.47% and 93.78%. These models, though slightly behind Ridge and Linear Regression, still showed strong predictive power. The MAE values for these models were 0.146663 (Gradient Boosting), 0.151654 (Lasso), and 0.167903 (Random Forest), indicating they made slightly larger average prediction errors compared to the top models.

At the lower end, AdaBoost and K-Nearest Neighbors (KNN) had R^2 scores of 92.63% and 89.02%, respectively, and showed a notable decline in performance. AdaBoost's MSE was 0.060884 with an MAE of 0.192798, while KNN had the highest MSE of 0.090838 and the largest MAE at 0.233688, indicating poor prediction accuracy and larger average errors for this dataset.

Overall, linear models like Ridge and Linear Regression performed the best, highlighting the linear relationship between the features and GPA. SVR and XGBoost showed strong results, but their added complexity may not be necessary given the success of simpler models. AdaBoost and KNN struggled due to their limitations in capturing underlying data relationships. In conclusion, Ridge Regression emerged as the best model, combining simplicity and accuracy for GPA prediction, with the lowest prediction errors (as evidenced by its low MSE and MAE values).

VI. CONCLUSION

The study compared several machine learning models for predicting student GPA, with the goal of identifying the most accurate and efficient model. Both Ridge Regression and Linear Regression achieved the highest R^2 score of 95.68%, showing strong predictive accuracy. Support Vector Regression (SVR) and XGBoost also performed well, achieving R^2 scores slightly lower than the linear models. Despite their complexity, these models did not substantially outperform Ridge or Linear Regression, which suggests that the relationship between features and GPA is mostly linear.

AdaBoost and K-Nearest Neighbors (KNN) were the weakest performers, with lower R^2 scores and higher prediction errors, indicating their unsuitability for this dataset. The study concluded that Ridge Regression is the most effective model due to its simplicity and high accuracy, while more complex models like SVR and XGBoost, though capable of handling non-linear relationships, do not offer enough improvement to justify their complexity.

Thus, Ridge Regression is recommended for GPA prediction, providing a balance between performance, interpretability, and simplicity, with Linear Regression serving as a strong alternative. Models such as KNN and AdaBoost should be avoided in this context due to their poor performance.

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